

LAB EXERCISE 5 – Looping Statements

Course Code: CCC1123 – Fundamental of C and Basic Input Output

Duration: 2 hours

Tasks

1. Write a program that accepts an integer N and uses a `for` loop to display all even numbers from 1 to N. Also display the total count.

Sample output :

```
Enter N: 20
Even numbers: 2 4 6 8 10 12 14 16 18 20
Total even numbers = 10
```

2. Ask the user for N and print the following pattern using both `for` statement.

```
*
**
***
****
*****
*****
*****
*****
*****
```

3. Using a `for` loop, calculate the factorial of a given integer N.
4. Write a program that keeps reading integers number using a `for` statement and calculate the average of the numbers.
5. Use a `for` loop to calculate the sum of all odd numbers between 1 and 100.
6. A supermarket introduces a Customer Reward Point System to encourage regular shopping. Each customer may purchase several items in a single visit, and points are given based on the total amount spent.

Reward Rules

Total Amount Spent (RM)	Reward Points
RM100 and above	50 points
RM50 – RM99	20 points
Below RM50	5 points

Sample Output

Enter number of items purchased : 4
Enter price of item 1 : 12.50
Enter price of item 2 : 8.40
Enter price of item 3 : 30.00
Enter price of item 4 : 15.00

Total amount spent : RM65.90
Reward points earned : 20
Thank you for shopping with us!

You are asked to write a program to help the supermarket calculate the total spending and determine the reward points for each customer. Apply input, selection, formula and for statement. (Hint: underline value enter by user)